

Green Multigene Genetic Programming: Bi-Objective Identification of Polynomial NARX Models under an Arithmetic-Cost Objective

Notas

- **O que mudou no MGGP:** o Green usa seleção de Pareto (NSGA-II), considerando erro e custo juntos. O modo MO-lex prioriza o erro e usa o custo para desempatar.
- **Correção comum:** no estudo novo, um avaliador externo corrige os atrasos de entrada e saída para todos os métodos. Mantém mínimos quadrados e os operadores; simulações que divergem são falhas, não zeros. O pacote original não foi editado.
- **Complexidade:** contamos as operações do polinômio sem termos repetidos: $C_\rho = a + \rho b$, com a adições e b multiplicações, incluindo os coeficientes. Isto é custo aritmético, não energia medida.
- **Dois pesos:** $\rho = 1$ conta todas as operações igualmente; $\rho_K = 729/64$ vem do modelo teórico em bits de Karatsuba. Os dois foram fixados antes das novas buscas e continuam na análise.
- **Medições:** tempos de execução permitem estimar $t_\times/t_+$; powermetrics mede potência para calcular energia; CodeCarbon fornece apenas a intensidade de carbono para calcular CO₂e. Energia e CodeCarbon não definem ρ .
- **Comparações novas:** além do protocolo original, todos os métodos receberam 4.510 avaliações por corrida. Também comparamos objetivos de número de termos e de nós, sem mudar a regra de escolha do modelo.
- **Estado deste documento:** os resultados agora usam o estudo corrigido, com orçamento igual e os dois pesos. A energia foi medida em uma nova sessão; os dados antigos foram preservados separadamente. Nem todas as reduções foram significativas, e os modelos escolhidos em E3 tiveram validação ruim.

1 Green MGGP: Formulation and Theory

This section formalises the model class, the multigene representation and the two objectives, establishes the properties on which Green MGGP relies, and characterises both the lexicographic scheme of the package and the proposed Pareto-based scheme. The main symbols are collected in Table 1, and Fig. 1 gives an overview of the method, from the data to the execution of the selected model, marking the components to which each theoretical result applies.

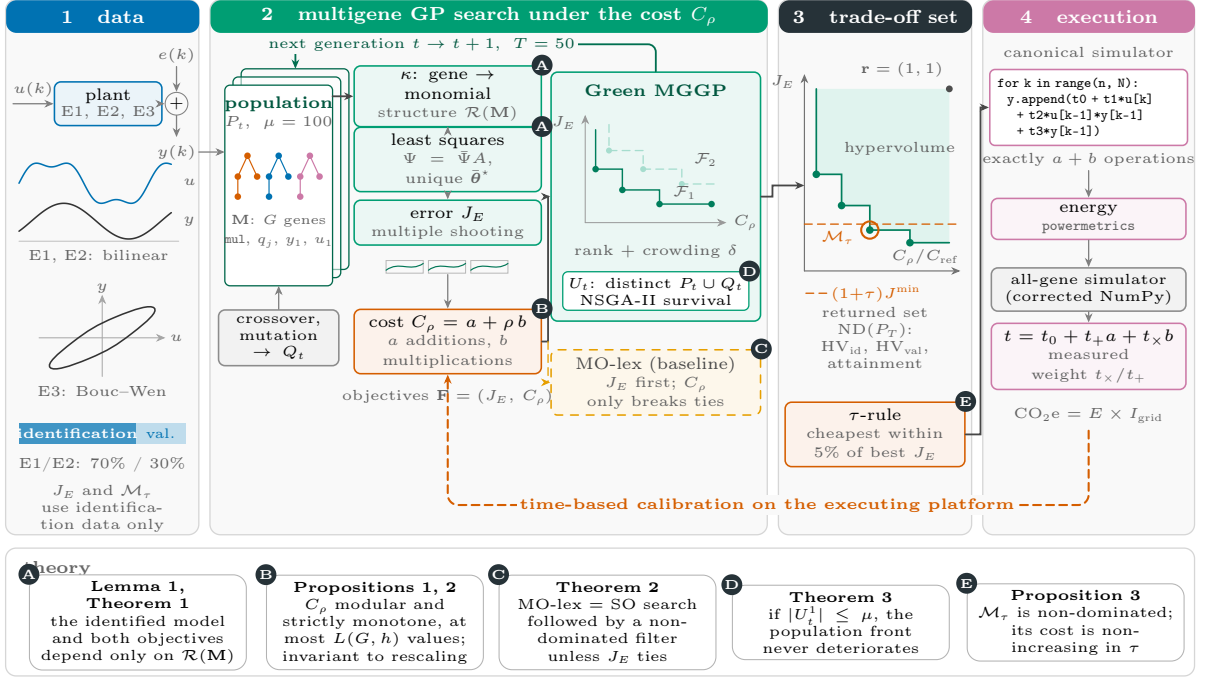


Figure 1: Overview of the methodology. (1) Identification and validation data of the three examples. (2) In Green MGGP, each individual of the population is mapped to its executed structure, estimated by least squares and evaluated by the multiple-shooting error J_E and the arithmetic cost C_ρ ; Pareto ranking with crowding and the removal of structural duplicates select the next population, whereas the MO-lex baseline compares the two objectives lexicographically. (3) The returned non-dominated set is assessed by its hypervolume and reduced to one model by the τ -rule. (4) The selected model is executed as a canonical polynomial, and execution time checks the relation between arithmetic cost and runtime; energy is measured separately. The dashed path denotes time-based calibration for a subsequent study, with its weight fixed before searching; it does not retune the reported searches. Letters A–E mark the components to which the theoretical results apply. Timing and energy are separate measurements within stage 4; the arrows indicate the execution-stage workflow. The 70%/30% split applies to E1/E2; E3 uses separate records.

Table 1: Main symbols.

Symbol	Meaning
$\mathbb{F} = \{y, u\} \times \mathbb{N}$	factors; (v, τ) denotes the delayed signal $k \mapsto v(k - \tau)$
$\mathfrak{M}, m $	monomials (finitely supported $m : \mathbb{F} \rightarrow \mathbb{N}$) and degree $ m = \sum_f m(f)$
φ_m, σ_j	regressor of m ; delay operator on monomials
$\mathcal{T}_h, \mathbf{M} = (g_1, \dots, g_G)$	genes of height at most h ; individual with G genes
$\kappa, \mathcal{R}(\mathbf{M})$	monomial semantics of a gene; executed structure of $\mathbf{M}$
$J_E, C_\rho, \mathbf{F} = (J_E, C_\rho)$	error objective, arithmetic cost, objective vector
$a, b, D = 2^h$	numbers of additions and multiplications, maximum degree
ρ, ρ_K	multiplication weight; bit-level (Karatsuba) weight $B^{\alpha-1}$
$\text{ND}(P), \text{HV}(P)$	non-dominated subset and hypervolume of a finite set P
μ, P_t, U_t	population size, population at generation t , de-duplicated union

1.1 Monomials and polynomial NARX structures

Polynomial NARX models express the output through monomials of delayed inputs and outputs [1]. The following definitions specify the representation used here. Let $\mathbb{N} = \{0, 1, \dots\}$ and let the input and output be sequences $u, y : \mathbb{Z} \rightarrow \mathbb{R}$. A factor $f = (v, \tau) \in \mathbb{F} = \{y, u\} \times \mathbb{N}$

represents the delayed signal $k \mapsto v(k - \tau)$.

Definition 1 (Monomials). A monomial is a finitely supported map $m : \mathbb{F} \rightarrow \mathbb{N}$; the set of monomials $\mathfrak{M}$ is the free commutative monoid generated by $\mathbb{F}$ under pointwise addition, with neutral element 0. Its degree is $|m| = \sum_{f \in \mathbb{F}} m(f)$, its maximum delay is $n(m) = \max\{\tau : m(v, \tau) > 0\}$, and its regressor is

$$\varphi_m(k) = \prod_{(v, \tau) \in \mathbb{F}} v(k - \tau)^{m(v, \tau)}. \quad (1)$$

For $j \in \mathbb{N}$, the delay operator $\sigma_j : \mathfrak{M} \rightarrow \mathfrak{M}$ is $(\sigma_j m)(v, \tau) = m(v, \tau - j)$ if $\tau \geq j$ and 0 otherwise.

Directly from the definition, $\varphi_{m+m'} = \varphi_m \varphi_{m'}$, $\varphi_{\sigma_j m}(k) = \varphi_m(k - j)$, $\sigma_i \sigma_j = \sigma_{i+j}$, $\sigma_j(m + m') = \sigma_j m + \sigma_j m'$ and $|\sigma_j m| = |m|$.

Definition 2 (Structure and model). A polynomial NARX structure is a finite non-empty set $\mathcal{R} \subset \mathfrak{M} \setminus \{0\}$, with maximum delay $n(\mathcal{R}) = \max_{m \in \mathcal{R}} n(m)$. For coefficients $\boldsymbol{\theta} = (\theta_0, (\theta_m)_{m \in \mathcal{R}})$, the model is

$$\hat{y}(k) = \theta_0 + \sum_{m \in \mathcal{R}} \theta_m \varphi_m(k), \quad (2)$$

which has $p = |\mathcal{R}| + 1$ terms.

1.2 Multigene representation and its semantics

Definition 3 (Genes and individuals). Let $\mathcal{T}$ be the set of finite trees generated by the grammar

$$g ::= y_1 \mid u_1 \mid \text{mul}(g, g) \mid q_j(g), \quad j \in \{1, \dots, n_q\},$$

and let $\mathcal{T}_h \subset \mathcal{T}$ contain the trees of height at most h , leaves having height 0. An individual is a G -tuple $\mathbf{M} = (g_1, \dots, g_G) \in \mathcal{T}_h^G$.

Multigene symbolic regression combines tree outputs through coefficients estimated by least squares [2]. Here we use the representation of the `mggp` package by Ávila dos Santos [3] with the multiplication primitive, where y_1 denotes $y(k - 1)$, u_1 denotes $u(k)$, `mul` is the pointwise product and q_j delays its argument by j samples. Every gene therefore denotes a monomial, as formalised next.

Definition 4 (Monomial semantics). The map $\kappa : \mathcal{T} \rightarrow \mathfrak{M}$ is defined recursively by $\kappa(y_1) = \mathbb{I}_{(y, 1)}$, $\kappa(u_1) = \mathbb{I}_{(u, 0)}$, $\kappa(\text{mul}(a, b)) = \kappa(a) + \kappa(b)$ and $\kappa(q_j(a)) = \sigma_j \kappa(a)$, where $\mathbb{I}_f$ is the indicator of the factor f . The *executed structure* of an individual is

$$\mathcal{R}(\mathbf{M}) = \{\kappa(g_1), \dots, \kappa(g_G)\}, \quad (3)$$

and two individuals are *structurally equivalent*, $\mathbf{M} \sim \mathbf{M}'$, if $\mathcal{R}(\mathbf{M}) = \mathcal{R}(\mathbf{M}')$.

Lemma 1 (Soundness and invariances of κ). *For every gene $g \in \mathcal{T}$:*

- (a) *the signal computed for g is the regressor $\varphi_{\kappa(g)}$;*
- (b) *κ is invariant under commutation and reassociation of `mul`, under composition of delays $q_i(q_j(a)) \mapsto q_{i+j}(a)$ and under distribution $q_j(\text{mul}(a, b)) \mapsto \text{mul}(q_j(a), q_j(b))$;*
- (c) *$|\kappa(g)|$ equals the number of leaves of g , so $1 \leq |\kappa(g)| \leq 2^h$ for $g \in \mathcal{T}_h$.*

Proof. (a) By structural induction. For the leaves, $\varphi_{\kappa(y,1)}(k) = y(k-1)$ and $\varphi_{\kappa(u,0)}(k) = u(k)$. If the claim holds for a and b , the product computed for $\text{mul}(a, b)$ is $\varphi_{\kappa(a)}\varphi_{\kappa(b)} = \varphi_{\kappa(a)+\kappa(b)}$, and the signal computed for $q_j(a)$ is $k \mapsto \varphi_{\kappa(a)}(k-j) = \varphi_{\sigma_j\kappa(a)}(k)$. (b) The four transformations are mapped by κ to the identities $m + m' = m' + m$, $(m + m') + m'' = m + (m' + m'')$, $\sigma_i\sigma_j = \sigma_{i+j}$ and $\sigma_j(m + m') = \sigma_j m + \sigma_j m'$ in $\mathfrak{M}$. (c) Each leaf contributes one unit of degree, mul adds degrees and σ_j preserves them. A tree of height h whose only binary node is mul has at most 2^h leaves. $\square$

Lemma 1(b) shows that many syntactically different individuals execute the same polynomial; for instance $\text{mul}(q_1(y_1), u_1)$ and $\text{mul}(u_1, q_1(y_1))$ both denote $y(k-2)u(k)$. The map $\mathbf{M} \mapsto \mathcal{R}(\mathbf{M})$ removes this redundancy, and $|\mathcal{R}(\mathbf{M})| \leq G$, with strict inequality when genes coincide. Fig. 2 illustrates these notions on an individual returned in a historical run, retained solely to illustrate the representation: its five genes execute only three monomials, so the polynomial that is actually evaluated, and whose operations define the cost introduced below, is smaller than the representation suggests.

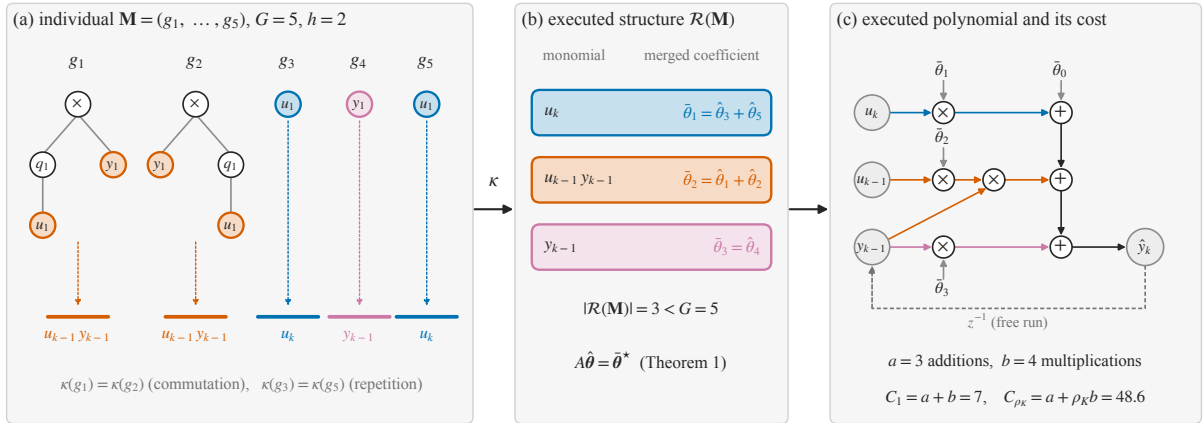


Figure 2: From an individual to its arithmetic cost, for a historical model returned by Green MGGP on E1 ($\rho = 1$, seed 13). (a) The five gene trees, with leaves coloured by the monomial that the gene denotes: g_1 and g_2 differ only by commutation, and g_3 and g_5 coincide. (b) The executed structure $\mathcal{R}(\mathbf{M})$ has three monomials, and the coefficients of genes with the same monomial are merged (Theorem 1). (c) The executed polynomial, simulated in free run, needs $a = 3$ additions and $b = 4$ multiplications per sample, so that $C_1 = 7$ and $C_{\rho_K} = 48.6$ in (5). This illustrative model omits the true bilinear regressor and is not the generating structure.

1.3 Parameter estimation and structural invariance

Let N identification samples be given and let $\mathcal{K} = \{n+1, \dots, N\}$, where n is the maximum delay of the individual. The regression matrix $\Psi(\mathbf{M}) \in \mathbb{R}^{|\mathcal{K}| \times (G+1)}$ has a column of ones and the columns $(\varphi_{\kappa(g_i)}(k))_{k \in \mathcal{K}}$, and $\bar{\Psi}(\mathcal{R}) \in \mathbb{R}^{|\mathcal{K}| \times (|\mathcal{R}|+1)}$ is the corresponding matrix with one column per element of $\mathcal{R}$.

Assumption 1 (Identifiability). $\bar{\Psi}(\mathcal{R})$ has full column rank.

Theorem 1 (Structural invariance of the identified model). *Let $\mathbf{M}$ be an individual with $\mathcal{R} = \mathcal{R}(\mathbf{M})$ satisfying Assumption 1, and let $A \in \{0, 1\}^{(|\mathcal{R}|+1) \times (G+1)}$ be the aggregation matrix with $A_{00} = 1$ and $A_{mi} = 1$ if and only if $\kappa(g_i) = m$. Then:*

- (i) $\Psi(\mathbf{M}) = \bar{\Psi}(\mathcal{R})A$;
- (ii) every minimiser $\hat{\theta}$ of $\|\mathbf{y} - \Psi(\mathbf{M})\theta\|_2$ satisfies $A\hat{\theta} = \bar{\theta}^*$, the unique minimiser of $\|\mathbf{y} - \bar{\Psi}(\mathcal{R})\bar{\theta}\|_2$;

(iii) the identified model of $\mathbf{M}$ coincides with the model (2) of $\mathcal{R}$ with coefficients $\bar{\boldsymbol{\theta}}^*$. Consequently its one-step-ahead, multiple-shooting and free-run predictions, and every objective computed from them, depend on $\mathbf{M}$ only through $\mathcal{R}(\mathbf{M})$.

Proof. (i) Column $i \geq 1$ of $\Psi(\mathbf{M})$ is the column of $\bar{\Psi}(\mathcal{R})$ indexed by $\kappa(g_i)$, that is, $\Psi e_i = \bar{\Psi} e_{\kappa(g_i)} = \bar{\Psi} A e_i$; the constant columns agree because $A_{00} = 1$. (ii) Every row of A contains a one, because every $m \in \mathcal{R}$ equals $\kappa(g_i)$ for some i , so A is surjective. Hence $\{\Psi\boldsymbol{\theta}\} = \{\bar{\Psi}A\boldsymbol{\theta}\} = \{\bar{\Psi}\bar{\boldsymbol{\theta}}\}$, and the two least-squares problems have the same optimal value. If $\hat{\boldsymbol{\theta}}$ is a minimiser of the first, $A\hat{\boldsymbol{\theta}}$ attains the optimal value of the second and is therefore a minimiser of it. Under Assumption 1 the second objective is strictly convex, so its minimiser is unique. (iii) For every argument, $\hat{\theta}_0 + \sum_{i=1}^G \hat{\theta}_i \varphi_{\kappa(g_i)} = \hat{\theta}_0 + \sum_{m \in \mathcal{R}} (\sum_{i: \kappa(g_i)=m} \hat{\theta}_i) \varphi_m = \theta_0^* + \sum_{m \in \mathcal{R}} \theta_m^* \varphi_m$. All three predictors evaluate this function recursively from the data and from initial conditions of length $n(\mathcal{R})$, which is itself a function of $\mathcal{R}$. $\square$

Corollary 1. *Every objective that is a function of the identified model and of $\mathcal{R}(\mathbf{M})$ is constant on the equivalence classes of $\sim$. Removing all but one member of each class from a set of individuals therefore removes no objective vector.*

1.4 Error objective

Multiple shooting limits simulation length by dividing the data into shorter segments [4]. Here the common evaluator resets each segment to measured outputs; this windowed criterion does not include the optimised segment initial states and continuity constraints of the formulation in [4].

Definition 5 (Multiple-shooting error). Let $n = n(\mathcal{R}(\mathbf{M}))$, $K \in \mathbb{N}$ and $W = \lfloor N/(n + K) \rfloor$. The identification samples are partitioned into W disjoint windows of length $n + K$. In window w , the identified model is initialised with the n measured outputs and simulated in free run for K steps, giving $\hat{y}_w(k)$. The error objective is

$$J_E(\mathbf{M}) = \frac{1}{\sigma_y} \left(\frac{1}{W(n + K)} \sum_{w=1}^W \sum_{k=1}^{n+K} (y_w(k) - \hat{y}_w(k))^2 \right)^{1/2}, \quad (4)$$

where σ_y is the standard deviation of the identification output, and $J_E(\mathbf{M}) = \infty$ if the estimation or the simulation fails.

The first n residuals of each window vanish by construction. By Theorem 1, J_E is a function of $\mathcal{R}(\mathbf{M})$. With $K = 20$ it penalises the accumulation of error over a finite horizon at a fraction of the cost of a full free-run simulation; the normalisation by σ_y does not affect the search.

A common evaluator in `corrected_experiments.py` applies to every configuration; the vendored package is unchanged. The evaluator builds least-squares columns with the intended factor delays, retains duplicate gene columns and aggregates their fitted coefficients for canonical simulation. It uses the maximum actual factor delay for initialisation, avoids circular indexing and treats non-finite predictions as failures rather than replacing them by zeros. These corrections also allow structures that the original predictor could not evaluate. They are applied to SO, MO-lex, Green and the ablations equally; historical caches are not reused.

1.5 Arithmetic-cost objective

Definition 6 (Arithmetic cost). Evaluating (2) in direct form requires $a(\mathcal{R}) = |\mathcal{R}|$ additions and $b(\mathcal{R}) = \sum_{m \in \mathcal{R}} |m|$ multiplications, one for each coefficient and $|m| - 1$ between the factors of each regressor. Let $\rho > 0$ be the cost of a multiplication relative to that of an addition on

the platform that executes the model. In addition-equivalents, the cost of a regressor and of an individual are

$$c_\rho(m) = 1 + \rho |m|, \quad C_\rho(\mathbf{M}) = \sum_{m \in \mathcal{R}(\mathbf{M})} c_\rho(m) = a(\mathcal{R}) + \rho b(\mathcal{R}). \quad (5)$$

Two weights are used. The bit-level weight of the Green-Box framework [5] charges B bit operations for an addition of B -bit operands and B^α , $\alpha = \log_2 3$, for a Karatsuba multiplication [6]; it corresponds to $\rho_K = B^{\alpha-1}$, which for binary64 operands ($B = 64$) equals $3^6/2^6 = 729/64 \approx 11.39$, and $B C_{\rho_K} = B a + B^\alpha b$ is the Green-Box complexity. The unit weight $\rho = 1$ makes $C_1 = a + b$ the number of arithmetic operations, which describes platforms on which both operations take comparable time, such as processors with floating-point units; Section 4 shows that it is the weight supported by the execution times measured in this study. We write C for C_ρ when the weight is clear.

1.5.1 Establishing the weights and separating the measurements

The two declared scenarios are analysed throughout: $\rho = 1$ is an unweighted operation count and $\rho_K = 729/64$ is a theoretical bit-level weight. Neither is obtained from power measurements. The bit-level model does not describe the instruction-level cost of binary64 multiplication on this laptop. Operation counts are computed from each canonical structure before evaluation, independently of any measured runtime, energy or carbon intensity.

A platform-specific timing calibration instead estimates $\rho_{\text{time}} = t_\times/t_+$ from $t = t_0 + t_+ a + t_\times b$. Here this fit characterises the Python simulator and includes interpreter overhead. Earlier timing measurements provided support for an order-one weight; they do not make $\rho = 1$ an exact fitted value. In the corrected study both declared weights were fixed before the new searches. Subsequent timing checks assess those choices and cannot retrospectively redefine their objectives. A newly calibrated weight would require a separately declared search protocol. Section 4 separates timing, power integration and carbon conversion; it reports the new measurements of the corrected selected models.

Proposition 1 (Structure of the cost). *Let $\rho > 0$ and let $\mathcal{R}, \mathcal{R}'$ be structures of individuals in $\mathcal{T}_h^G$, with $D = 2^h$:*

- (i) C_ρ is modular: $C_\rho(\mathcal{R} \cup \mathcal{R}') = C_\rho(\mathcal{R}) + C_\rho(\mathcal{R}') - C_\rho(\mathcal{R} \cap \mathcal{R}')$;
- (ii) C_ρ is strictly monotone: $\mathcal{R} \subsetneq \mathcal{R}'$ implies $C_\rho(\mathcal{R}) < C_\rho(\mathcal{R}')$, and $|m| < |m'|$ implies $c_\rho(m) < c_\rho(m')$;
- (iii) $(1 + \rho) a(\mathcal{R}) \leq C_\rho(\mathcal{R}) \leq (1 + \rho D) a(\mathcal{R})$;
- (iv) C_ρ depends on $\mathcal{R}$ only through (a, b) , and the number of distinct values of C_ρ over $\mathcal{T}_h^G$ is at most

$$L(G, h) = \sum_{a=1}^G (a(D-1) + 1) = G + \frac{(D-1)G(G+1)}{2}; \quad (6)$$

- (v) for $\rho = \rho_K$ and $G < 729$, the map $(a, b) \mapsto C_\rho$ is injective, so the cost determines both the number of terms and the total degree; for $\rho = 1$ it is not, and C_1 takes at most $G(D+1) - 1$ values.

Proof. (i) C_ρ is a sum of c_ρ over the elements of a set. (ii) $c_\rho(m) > 0$ for every m , and c_ρ is strictly increasing in $|m|$. (iii) By Lemma 1(c), $1 \leq |m| \leq D$. (iv) The first statement is (5); moreover $a \in \{1, \dots, G\}$ and, by (iii), $a \leq b \leq aD$, which gives at most $a(D-1) + 1$ values of b for each a . (v) $\rho_K = 64^{\log_2 3 - 1} = 3^6/2^6$. If $a + \rho_K b = a' + \rho_K b'$, then $64(a - a') = 729(b' - b)$; since $\gcd(64, 729) = 1$, 729 divides $a - a'$, and $|a - a'| < G < 729$ forces $a = a'$ and then $b = b'$.

For $\rho = 1$, $(a, b) = (2, 2)$ and $(1, 3)$ have the same cost, and $C_1 = a + b$ ranges over the integers from 2 to $G(D + 1)$. $\square$

Remark 1. The complexity used in the tutorial of the package, ω times the number of `mul` nodes, assigns $c_\omega(m) = \omega(|m| - 1)$ to each gene and sums over genes rather than over $\mathcal{R}(\mathbf{M})$. It violates Proposition 1(ii), because a linear regressor costs nothing, so adding linear terms never increases the cost, and it charges duplicated genes that are executed once.

Proposition 2 (Invariance to monotone rescaling). *Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be strictly increasing. The Pareto order and the lexicographic order induced by $(J_E, \psi \circ C)$ coincide with those induced by (J_E, C) . In particular, any constant factor $\omega > 0$ applied to a cost has no effect on an algorithm whose decisions depend only on these orders, and the Green-Box complexity BC_{ρ_K} induces the same orders as C_{ρ_K} .*

Proof. ψ preserves $\leq$, $<$ and $=$, and both orders are defined from these relations applied componentwise. $\square$

The weight ρ , in contrast, is not a rescaling: it changes the relative cost of structures with different numbers of terms and degrees, and hence the orders. A structure with fewer, higher-degree terms can be cheaper than one with more, linear terms under $\rho = 1$ and more expensive under ρ_K , so the weight must reflect the target platform.

1.6 Bi-objective problem

We use Pareto dominance to compare the objectives and hypervolume to measure the dominated region [7].

Definition 7 (Dominance, Pareto set, hypervolume). For $\mathbf{F} = (J_E, C)$, $\mathbf{M}$ dominates $\mathbf{M}'$, written $\mathbf{M} \prec \mathbf{M}'$, if $\mathbf{F}(\mathbf{M}) \leq \mathbf{F}(\mathbf{M}')$ componentwise and $\mathbf{F}(\mathbf{M}) \neq \mathbf{F}(\mathbf{M}')$. For a finite set P , $\text{ND}(P)$ is its subset of non-dominated elements. The problem is

$$\min_{\mathbf{M} \in \mathcal{T}_h^G} \mathbf{F}(\mathbf{M}) = (J_E(\mathbf{M}), C(\mathbf{M})), \quad (7)$$

whose solution is the set of non-dominated individuals. For a reference point $\mathbf{r}$, the hypervolume of P is the Lebesgue measure $\text{HV}(P) = \lambda(\bigcup_{\mathbf{M} \in P} [\mathbf{F}(\mathbf{M}), \mathbf{r}])$.

If every element of P is weakly dominated by an element of Q , then the region dominated by P is contained in that dominated by Q , so $\text{HV}(P) \leq \text{HV}(Q)$; this monotonicity is used below.

1.7 Selection schemes

Both schemes share one generation template: parent selection by binary tournament, crossover and mutation with the operators of the package, common least-squares estimation and predictor evaluation, and elitist survival. The equal-evaluation protocol evaluates every offspring, including unchanged copies. They differ only in the relation used to compare individuals and in the survival rule.

Lexicographic scheme (MO-lex). In the bi-objective mode of the package [3], tournaments and the hall of fame use DEAP fitness comparisons [8, 9], which follow the lexicographic order: $\mathbf{F}(\mathbf{M}) <_{\text{lex}} \mathbf{F}(\mathbf{M}')$ if $J_E(\mathbf{M}) < J_E(\mathbf{M}')$, or $J_E(\mathbf{M}) = J_E(\mathbf{M}')$ and $C(\mathbf{M}) < C(\mathbf{M}')$. The single-objective configuration (SO) compares J_E only. The run returns $\text{ND}(P_T)$ in MO-lex and the best individual in SO, where P_T is the final population.

Algorithm 1 One generation of Green MGGP

Require: population P_t with $|P_t| = \mu$; offspring count λ ; crossover and mutation rates p_c, p_m

- 1: assign rank and crowding distance (8) to all $\mathbf{M} \in P_t$
- 2: $Q_t \leftarrow \lambda$ copies selected by binary tournament under the crowded order
- 3: apply package crossover to consecutive pairs of Q_t with probability p_c
- 4: apply package mutation to each $\mathbf{M} \in Q_t$ with probability p_m
- 5: estimate $\hat{\theta}$ and evaluate $\mathbf{F}$ for every offspring in the equal-evaluation protocol
- 6: $U_t \leftarrow$ one representative per class of $\sim$ among the elements of $P_t \cup Q_t$ with finite $\mathbf{F}$
- 7: $P_{t+1} \leftarrow$ the first μ elements of U_t under the crowded order (NSGA-II truncation); if $|U_t| < \mu$, complete with the remaining individuals

Theorem 2 (Lexicographic reduction). *Run SO and MO-lex from the same initial population with the same pseudo-random stream. Suppose that, up to generation t , every pair of individuals compared by a tournament or by the hall of fame with $J_E(\mathbf{M}) = J_E(\mathbf{M}')$ also satisfies $C(\mathbf{M}) = C(\mathbf{M}')$. Then both runs produce the same populations $P_0, \dots, P_t$. If the condition holds for all generations, MO-lex returns $\text{ND}(P_t^{\text{SO}})$, the non-dominated subset of the final SO population.*

Proof. By induction on the generation. P_0 is common. Suppose that P_s is common. A generation is a deterministic function of the current population, of the pseudo-random numbers it consumes and of the outcomes of the comparisons it performs; the number and order of the random draws depend only on the population and on these outcomes. For a compared pair with $J_E(\mathbf{M}) \neq J_E(\mathbf{M}')$, both relations order the pair as J_E does. For a pair with $J_E(\mathbf{M}) = J_E(\mathbf{M}')$, the hypothesis gives $C(\mathbf{M}) = C(\mathbf{M}')$, so both relations declare a tie, which the implementation resolves by position in the same way. All outcomes coincide, the same random numbers are consumed, the same offspring are produced and evaluated deterministically, and P_{s+1} is common. The adaptation of the mutation rate depends on J_E of the best individual and is therefore also common. The last statement follows from $P_T^{\text{lex}} = P_T^{\text{SO}}$. $\square$

Theorem 2 states that the lexicographic scheme is the single-objective algorithm followed by a non-dominated filter, unless exact ties in J_E between individuals with different costs occur. Equivalent structures and failed evaluations can produce exact ties; continuity alone does not exclude them. Section 3 checks the population trajectories empirically.

Pareto-based scheme (Green MGGP). Green MGGP replaces the comparison relation and the survival rule by those of NSGA-II [10].

Definition 8 (Rank and crowding). For a finite set P , let $\mathcal{F}_1 = \text{ND}(P)$ and $\mathcal{F}_{r+1} = \text{ND}(P \setminus \bigcup_{s \leq r} \mathcal{F}_s)$. The rank of $\mathbf{M} \in \mathcal{F}_r$ is r , and its crowding distance is

$$\delta(\mathbf{M}) = \sum_{i=1}^2 \frac{F_i(\mathbf{M}_i^+) - F_i(\mathbf{M}_i^-)}{F_i^{\max} - F_i^{\min}}, \quad (8)$$

where $\mathbf{M}_i^\pm$ are the neighbours of $\mathbf{M}$ in $\mathcal{F}_r$ sorted by F_i , the extremes and normalisation are taken over $\mathcal{F}_r$, and boundary elements receive $\delta = \infty$. The crowded order prefers lower rank and, for equal ranks, larger δ .

Theorem 3 (Non-deterioration of the population front). *Let $U_t^1 = \text{ND}(U_t)$ in Algorithm 1 and suppose that P_t contains an individual with finite $\mathbf{F}$. If $|U_t^1| \leq \mu$, then:*

- (i) $U_t^1 \subseteq P_{t+1}$;
- (ii) for every $\mathbf{M} \in \text{ND}(P_t)$ there is $\mathbf{M}' \in \text{ND}(P_{t+1})$ with $\mathbf{F}(\mathbf{M}') \leq \mathbf{F}(\mathbf{M})$;

(iii) $\text{HV}(\text{ND}(P_{t+1})) \geq \text{HV}(\text{ND}(P_t))$ for every reference point.

Moreover, if no two elements of U_t^1 share an objective vector, then $|U_t^1| \leq L(G, h)$ in (6), so $L(G, h) \leq \mu$ is a sufficient condition for (i)–(iii) at every generation.

Proof. (i) NSGA-II truncation admits the fronts of U_t in increasing rank, and $\mathcal{F}_1 = U_t^1$ fits entirely. (ii) Let $\mathbf{M} \in \text{ND}(P_t)$. Since P_t has an element with finite $\mathbf{F}$, so does $\mathbf{M}$. Its class has a representative $\tilde{\mathbf{M}} \in U_t$ with $\mathbf{F}(\tilde{\mathbf{M}}) = \mathbf{F}(\mathbf{M})$ (Corollary 1). Because U_t is finite and $\prec$ is a strict partial order, either $\tilde{\mathbf{M}} \in U_t^1$ or some $\mathbf{M}' \in U_t^1$ dominates $\tilde{\mathbf{M}}$; in both cases there is $\mathbf{M}' \in U_t^1$ with $\mathbf{F}(\mathbf{M}') \leq \mathbf{F}(\mathbf{M})$, and $\mathbf{M}' \in P_{t+1}$ by (i). Every element of P_{t+1} either belongs to U_t , shares its objective vector with an element of U_t , or has $\mathbf{F} = (\infty, \infty)$; none of them dominates $\mathbf{M}'$, so $\mathbf{M}' \in \text{ND}(P_{t+1})$. (iii) follows from (ii) and the monotonicity of the hypervolume. For the bound, two elements of a non-dominated set with the same cost must have the same error, since otherwise one dominates the other; when no two elements share an objective vector, U_t^1 has at most one element per value of C_ρ , and Proposition 1(iv) bounds the number of these values by $L(G, h)$ for every $\rho > 0$. $\square$

The single-objective and lexicographic schemes enjoy no such guarantee for the cost dimension: their elitism retains the individuals with the lowest J_E , so a cheap non-dominated individual can be lost from one generation to the next.

1.8 Selecting a single model

Definition 9 (τ -rule). For a returned set P , $\tau \geq 0$ and $\eta \geq 0$, let $J^{\min} = \min_{\mathbf{M} \in P} J_E(\mathbf{M})$ and

$$\mathcal{M}_\tau \in \arg \min \{ C(\mathbf{M}) : \mathbf{M} \in P, J_E(\mathbf{M}) \leq (1 + \tau)J^{\min} + \eta \}, \quad (9)$$

with ties broken by the smaller J_E .

Proposition 3 (Properties of the τ -rule). (i) $\mathcal{M}_\tau \in \text{ND}(P)$. (ii) $\tau \mapsto C(\mathcal{M}_\tau)$ is non-increasing; if $P = \text{ND}(P)$, then $\tau \mapsto J_E(\mathcal{M}_\tau)$ is non-decreasing. (iii) $\mathcal{M}_0$ is the cheapest model of minimal error up to η , and for τ large enough $\mathcal{M}_\tau$ is the cheapest model in P .

Proof. (i) Suppose $\mathbf{M} \prec \mathcal{M}_\tau$. Then $J_E(\mathbf{M}) \leq J_E(\mathcal{M}_\tau)$, so $\mathbf{M}$ is feasible, and $C(\mathbf{M}) \leq C(\mathcal{M}_\tau)$. Minimality of the cost forces $C(\mathbf{M}) = C(\mathcal{M}_\tau)$, and then $J_E(\mathbf{M}) < J_E(\mathcal{M}_\tau)$ contradicts the tie-breaking rule. (ii) The feasible set grows with τ , so its minimum cost cannot increase. If P is non-dominated, a smaller cost implies a larger error, and an equal cost implies an equal vector. (iii) is immediate from the definition. $\square$

We use $\tau = 0.05$ and $\eta = 10^{-6}$, fixed before the experiments; $\mathcal{M}_\tau$ is then the cheapest model whose identification error lies within 5% of the best error found, and η only matters on noise-free data, where the best error is at machine precision. For SO, P contains a single model. The rule uses identification data only, and its sensitivity to τ is examined in Section 3.

2 Experimental Setup

2.1 Examples

The study uses the regression examples distributed with the `mggp` package [3] (Table 2, Fig. 3); the classification mode of the package returns a single objective and is not considered. Example E1 is the tutorial system

$$y(k) = 0.65 y(k-1) + 0.35 u(k) + 0.10 u(k) y(k-1), \quad (10)$$

excited by $u = \sin t + 0.25 \sin 3t$ over 300 samples, of which the first 70% are used for identification and the remainder for validation. Example E2 adds white Gaussian measurement noise with

standard deviation 0.02 to the output of E1. Example E3 uses the package simulator of the Bouc–Wen hysteresis model [11], $y = d_p u - h$ with $\dot{h} = A\dot{u} - \beta|\dot{u}|h - \gamma\dot{u}|h|$, $A = 0.9$, $\beta = \gamma = 0.008$ and $d_p = 1.6$, integrated by fourth-order Runge–Kutta with a step of 1 ms. Identification uses a 1 Hz sinusoid whose amplitude grows linearly from 20 to 50 over 6 s, and validation a 0.8 Hz sinusoid of amplitude 45 over 4 s; both are decimated by 20.

Table 2: Examples and search settings; C_{ref} for $\rho = 1$ (common: population 100, 50 generations, crossover 0.9, initial mutation 0.2; baseline elite retention 10%, multiple-shooting horizon $K = 20$, primitives `mul` and q^{-j}).

Example	N_{id}	N_{val}	genes	height	j_{max}	C_{ref}
E1 Tutorial (noise-free)	210	90	5	2	2	25
E2 Tutorial (noisy)	210	90	5	2	2	25
E3 Bouc–Wen hysteresis	300	200	8	3	3	72

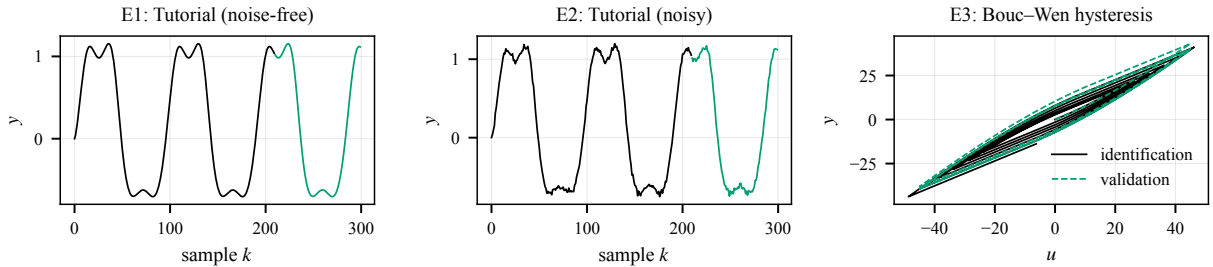


Figure 3: Identification and validation data of the three examples. E3 is shown in the input–output plane, where the loops reveal the hysteresis.

2.2 Common evaluator, selection schemes and search budgets

SO minimises identification error and delivers its Hall-of-Fame model. MO-lex uses the arithmetic cost as a second fitness component but compares it lexicographically; Green uses Pareto ranking and crowding. All methods use the external corrected evaluator described in Section 1, the same examples, least-squares estimator, representation, variation operators and initial rates. Both declared weights are fixed before searching: $\rho = 1$ is primary and $\rho_K = 729/64$ is the bit-level comparison. SO does not use either cost for selection.

The main comparison uses 100 initial evaluations followed by 49 steps of 90 evaluated offspring, giving exactly 4,510 fitness calls per run, including unchanged offspring and failures. The 50 recorded generations include the initial population. The initial mutation probability is 0.2; at nine-generation checkpoints it increases by 0.1, up to 0.9, when improvement is below 5%. Baseline survival retains 10% elites, whereas Pareto configurations use union-based survival.

Two additional Pareto configurations minimise distinct canonical terms or all representation nodes, respectively, alongside identification error. Node counting includes duplicate genes; structural de-duplication retains the smallest representation for this ablation. The same arithmetic cost and identification-only τ -rule assess the delivered sets of every configuration. Each of the five configurations uses 30 independent seeds on each example under both weights (900 runs). A separate native-budget diagnostic retains the original offspring counts and fitness reuse for SO, MO-lex and Green (540 runs); it is not the principal equal-evaluation comparison. The complete study therefore contains 1,440 runs.

2.3 Assessment and statistical comparisons

Hypervolume [7] is computed in $(C_\rho/C_{\text{ref}}, \text{NRMSE})$ with reference $(1, 1)$ and the cost bound declared from G genes of degree $2^{h_{\text{max}}}$. Identification and free-run validation hypervolumes assess the same delivered models. Points outside the reference contribute nothing; reference-point sensitivity is analysed separately [12]. The fixed $\tau = 0.05$, $\eta = 10^{-6}$ rule uses identification data only. Validation initialises the actual maximum delay from measured outputs; non-finite simulations remain failures and are excluded from finite-error medians, with their counts retained.

Comparisons use two-sided Mann–Whitney tests [13], Holm correction [14] and Vargha–Delaney A_{12} [15]. Seeds index independent runs; method comparisons are not paired. Holm families contain 60 tests per weight in the equal-evaluation arm and 30 in the native diagnostic, covering the tested metrics, examples and comparator configurations. Runtime recorded during concurrent searches is diagnostic; the isolated timing and energy session is reported separately in Section 4.

3 Results and Discussion

The following results use the corrected common evaluator and equal-evaluation protocol. The primary weight is $\rho = 1$; Section 3.3 reports the Karatsuba-weighted scenario. All values come from the new caches, not the historical experiment.

Table 3: Results over 30 independent runs with $\rho = 1$: median [interquartile range]. HV: hypervolume of the delivered set (identification or free-run validation error); C_τ , NRMSE_τ : cost and free-run validation NRMSE of the model selected by the τ -rule; Rec_τ , Rec_p : fraction of runs in which the selected model, or at least one model of the returned set, contains the three true regressors of (10). $^*/^\dagger$: Green differs from SO/MO-lex (Mann–Whitney, Holm-adjusted $p < 0.05$).

Ex.	Method	HV _{id}	HV _{val}	C_τ	NRMSE_τ	Rec_τ	Rec_p
E1	SO	0.640 [0.560, 0.640]	0.640 [0.560, 0.640]	9 [9, 11]	3e-16 [2e-16, 3e-16]	1.00	1.00
	MO-lex	0.829 [0.803, 0.829]	0.827 [0.798, 0.827]	7 [7, 7]	3e-16 [3e-16, 3e-16]	1.00	1.00
	Green	0.896 [0.896, 0.896] ^{*†}	0.893 [0.893, 0.893] ^{*†}	7 [7, 7] [*]	9e-16 [7e-16, 2e-15] ^{*†}	1.00	1.00
E2	SO	0.508 [0.508, 0.508]	0.505 [0.505, 0.505]	12 [12, 12]	0.028 [0.028, 0.028]	0.00	0.00
	MO-lex	0.702 [0.661, 0.703]	0.699 [0.656, 0.700]	10 [9, 10]	0.028 [0.028, 0.029]	0.00	0.07
	Green	0.878 [0.878, 0.879] ^{*†}	0.872 [0.872, 0.874] ^{*†}	7 [7, 7] ^{*†}	0.028 [0.028, 0.028]	0.00	0.00
E3	SO	0.736 [0.729, 0.770]	0.000 [0.000, 0.000]	18.5 [16, 19]	1.394 [1.301, 1.463]	–	–
	MO-lex	0.798 [0.797, 0.825]	0.000 [0.000, 0.471]	16 [16, 16]	1.411 [1.340, 1.467]	–	–
	Green	0.955 [0.932, 0.956] ^{*†}	0.893 [0.868, 0.893] ^{*†}	16 [14.5, 16] [*]	1.420 [1.316, 1.451]	–	–

3.1 Delivered trade-off sets and selected models

Green achieves median identification hypervolumes of 0.896, 0.878 and 0.955 on E1–E3, respectively, compared with 0.829, 0.702 and 0.798 for MO-lex. Its corresponding validation hypervolumes are 0.893, 0.872 and 0.893. These indicators assess delivered sets: SO delivers one model, while MO-lex and Green deliver sets. They therefore do not establish an advantage for every selected model or every restricted high-accuracy region. Figure 5 locates these differences along the cost axis rather than averaging them into a single hypervolume [16]. Each step gives the median, over runs, of the smallest error among delivered models whose cost does not exceed the horizontal coordinate; the band shows the interquartile range. Moving down means better accuracy, and reaching the same level further left means lower arithmetic cost. The upper row uses identification error and the lower row free-run validation of the same models.

On E1, the curves reach errors near machine precision, so the relevant distinction is how cheaply a method reaches effectively exact prediction, rather than differences between tiny residuals. E2 shows that Green offers low-cost alternatives without reproducing the generating structure. E3 has a larger separation between identification and validation: reducing identification error along a front does not ensure improved free-run prediction. In particular, good validation points in a delivered set need not be the points retained by the identification-only τ -rule. The set-level curves and selected-model errors therefore answer different questions. The

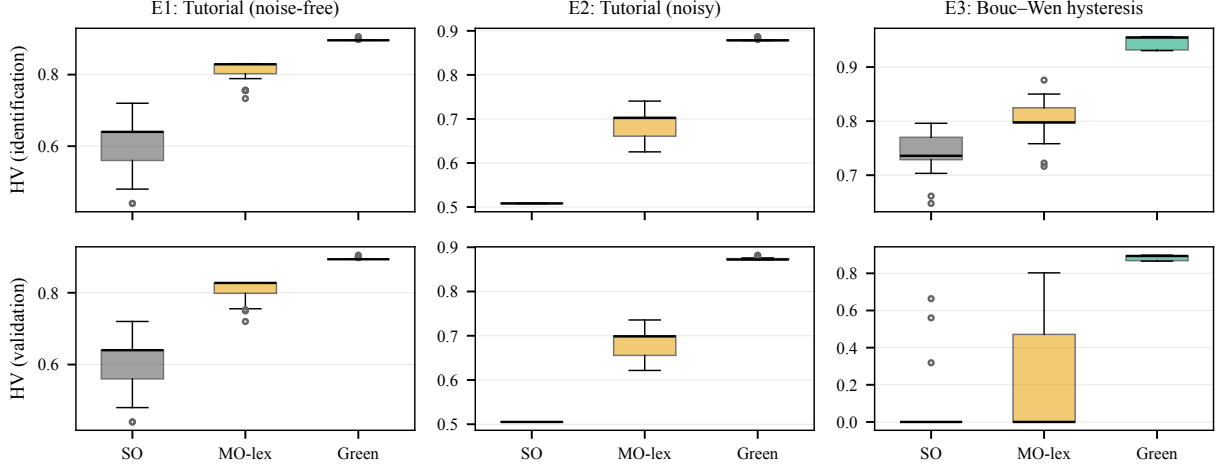


Figure 4: Corrected equal-evaluation study: identification and validation hypervolumes of the delivered sets over 30 runs with $\rho = 1$.

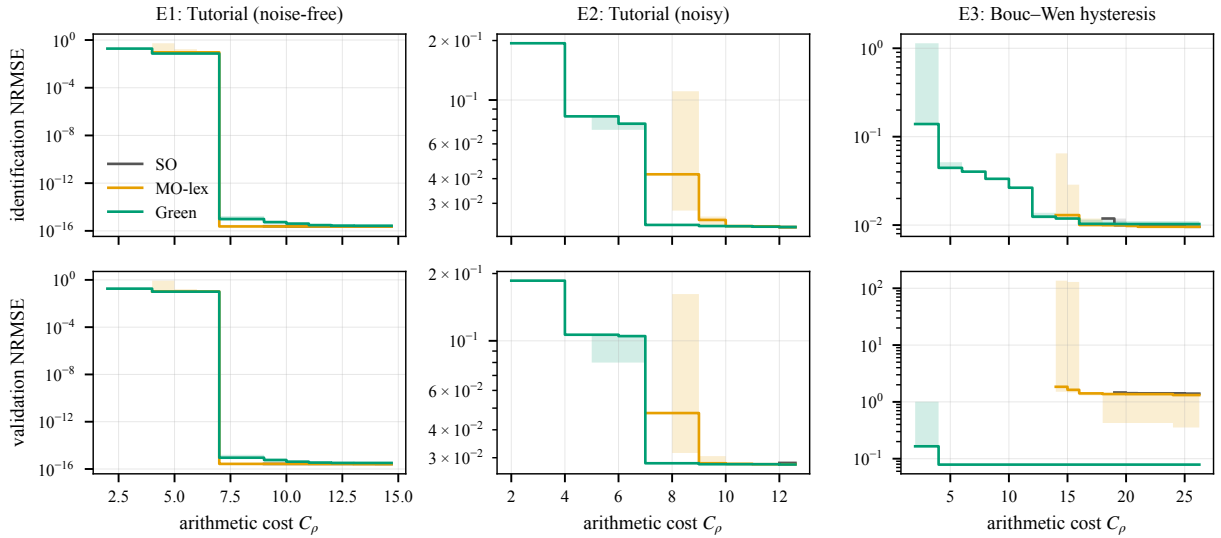


Figure 5: Median attainment curves and interquartile bands of the delivered sets in the corrected study.

selected Green costs are 7, 7 and 16 operations, against 9, 12 and 18.5 for SO. On E1 all selected models are effectively exact; statistical differences below 10^{-14} have no practical accuracy interpretation. On noisy E2 the selected models do not recover the generating regressors, and the corrected Green sets do not retain the historical structure-recovery result. The earlier claim of full E2 recovery therefore does not apply to this study.

On E3, selection based only on identification error gives poor validation. The selected Green NRMSE is 1.420, compared with 1.394 for SO and 1.411 for MO-lex. Green sets contain models with substantially better validation, but the fixed τ -rule does not choose them. Neither the

rule nor its parameters are changed after observing validation. These outcomes prevent an accuracy-preserving interpretation of the E3 selected-model energy comparison. Table 4 makes the structural differences concrete for the declared representative runs. On E2, Green and MO-lex select the same three delayed regressors, whereas SO retains five. Similar validation errors therefore coexist with different operation counts. None of these representative structures is the three-regressor generating structure: prediction accuracy on the available record does not establish recovery of the original delays.

On E3, the representative Green model consists of seven linear delayed terms; the baselines also include a cubic product. Counting factors and coefficient multiplications distinguishes these structures more directly than counting terms alone. The representative Green model has a different cost from the across-run median because the representative is chosen by median validation NRMSE, not median cost. All three representative E3 models have poor free-run validation, consistent with the limitation of the fixed selection rule.

Table 4: Regressors of the model selected by the τ -rule in the representative run (median validation NRMSE of the selected model) and its free-run validation NRMSE.

Ex.	Method	Regressors	C	NRMSE
E2	SO	$u(k)y(k-2), u(k-1), y(k-2), y(k-2)^2, y(k-5)$	12	0.028
E2	MO-lex	$u(k-1), u(k-1)y(k-2), y(k-5)$	7	0.028
E2	Green	$u(k-1), u(k-1)y(k-2), y(k-5)$	7	0.028
E3	SO	$u(k-2)u(k-4)y(k-3), u(k-6), u(k-7), u(k-8), y(k-1), y(k-2), y(k-7), y(k-8)$	18	1.413
E3	MO-lex	$u(k-3)u(k-4)y(k-2), u(k-6), u(k-7), u(k-8), y(k-1), y(k-2), y(k-8)$	16	1.418
E3	Green	$u(k), u(k-1), u(k-2), u(k-3), y(k-1), y(k-7), y(k-8)$	14	1.425

3.2 Term and node ablations

Table 5: Corrected equal-evaluation ablation: medians over 30 runs with $\rho = 1$.

example	config	HV (id)	HV (val)	Cost	NRMSE	Evaluations
E1	Green	0.896	0.893	7.000	0.000	4510.000
	MO-lex	0.829	0.827	7.000	0.000	4510.000
	Pareto-nodes	0.831	0.828	7.000	0.000	4510.000
	Pareto-terms	0.896	0.893	7.000	0.000	4510.000
	SO	0.640	0.640	9.000	0.000	4510.000
E2	Green	0.878	0.872	7.000	0.028	4510.000
	MO-lex	0.702	0.699	10.000	0.028	4510.000
	Pareto-nodes	0.812	0.807	11.000	0.028	4510.000
	Pareto-terms	0.878	0.872	7.000	0.028	4510.000
	SO	0.508	0.505	12.000	0.028	4510.000
E3	Green	0.955	0.893	16.000	1.420	4510.000
	MO-lex	0.798	0.000	16.000	1.411	4510.000
	Pareto-nodes	0.929	0.868	14.000	1.425	4510.000
	Pareto-terms	0.954	0.893	16.000	1.189	4510.000
	SO	0.736	0.000	18.500	1.394	4510.000

The term-count ablation has no significant difference in validation hypervolume from Green on any tested example after Holm correction. Thus these experiments do not establish a con-

sistent advantage of the arithmetic objective over term count, although arithmetic counting explicitly distinguishes additions and multiplications. The node objective assesses the full representation rather than the duplicate-free polynomial. Finite validation medians in the ablation table exclude failures: on E3, six term-ablation runs, one SO run and one MO-lex run fail validation under $\rho = 1$. A smaller median among the remaining term-ablation runs must be interpreted alongside those failures.

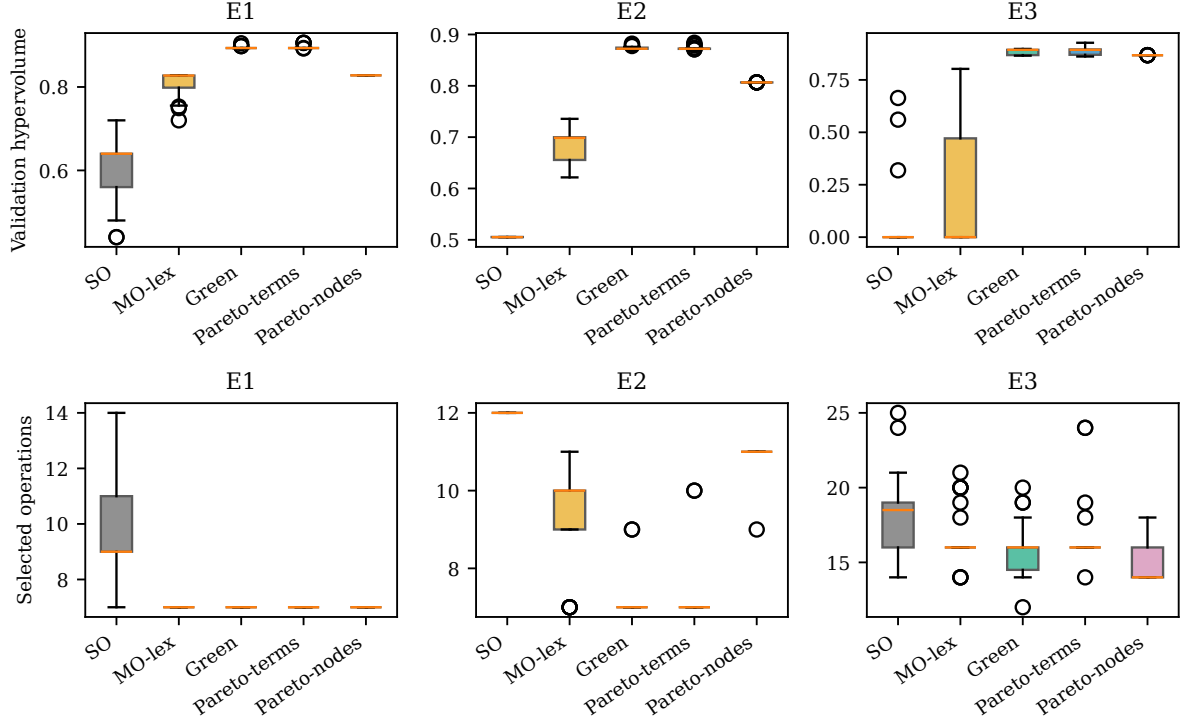


Figure 6: Equal-evaluation comparison with term and node objectives: validation hypervolume and selected arithmetic cost over 30 runs.

3.3 Multiplication-weight comparison

Table 6: Effect of the multiplication weight ρ over 30 runs (medians). HV: identification hypervolume; ΔC_τ : reduction of the median selected cost of Green MGGP relative to SO, with the cost of the same scenario; $a+b$ and C_{ρ_K} : operation count and Karatsuba-weighted cost of the selected Green model; same: fraction of seeds in which Green MGGP selects the same structure under both weights.

Ex.	weight	HV _{id}			ΔC_τ	Green $\mathcal{M}_\tau$		
		SO	MO-lex	Green		$a+b$	C_{ρ_K}	same
E1	$\rho = 1$	0.640	0.829	0.896	22%	7	48.6	100%
	ρ_K	0.738	0.884	0.929	20%	7	48.6	
E2	$\rho = 1$	0.508	0.702	0.878	42%	7	48.6	80%
	ρ_K	0.622	0.773	0.910	43%	7	48.6	
E3	$\rho = 1$	0.736	0.798	0.955	14%	16	109.5	0%
	ρ_K	0.819	0.874	0.969	14%	16	109.5	

Both weights remain in the analysis. Table 6 reports the separate cost objectives and the

fraction of identical selected Green structures. Changing ρ changes relative structure costs, rather than merely rescaling them. Cost reductions in this table are reductions between method medians, not medians of per-run reductions; hypervolumes under different weights use different cost coordinates and cannot be treated as the same indicator. The weights were fixed before the corrected searches and are assessed by timing measurements in Section 4.

3.4 Theory and implementation checks

Table 7: Empirical verification of Theorems 2 and 3 over 30 seeds. Lexicographic reduction: seeds in which SO and MO-lex produce identical populations in all generations; diverging seeds in which the divergence is preceded by an error tie between individuals of different cost, and in which the populations involved contain at least two individuals with $J_E = \infty$; median generation of the first divergence. Non-deterioration: largest $|U_t^1|$ over all runs and generations, bound $L(G, h)$ and population size μ ; runs with at least one decrease of the population-front hypervolume.

Ex.	Lexicographic reduction				Capacity		Runs with HV decrease			
	identical	tie	$J_E = \infty$	first div.	$\max U_t^1 $	$L(G, h)$	μ	SO	MO-lex	Green
E1	30/30	0/0	0/0	–	8	50	100	30	30	0
E2	30/30	0/0	0/0	–	9	50	100	30	30	0
E3	30/30	0/0	0/0	–	10	260	100	30	30	0

Table 7 separates the two theoretical checks. Population hashes show identical SO and MO-lex trajectories in all 30 runs on each example for the primary weight, consistent with Theorem 2; their delivered sets differ through final filtering. Green shows no population-front hypervolume decreases, and the observed union-front sizes satisfy the capacity condition of Theorem 3. These checks support the conditional theoretical statements rather than proving them for arbitrary searches. Canonical regression matrices of all 450 selected models across the five primary-weight configurations have full numerical column rank; this does not check every candidate evaluated during search.

4 Execution Time, Energy and Carbon of the Corrected Models

This separate stage reads the equal-evaluation caches and measures execution of their selected models. It does not alter C_ρ , either declared weight or the τ -rule. Historical energy measurements are not included.

4.1 Reconstruction and execution implementations

The 450 selected records comprise SO, MO-lex and Green under $\rho = 1$, and MO-lex and Green under ρ_K , across three examples and 30 seeds. Four failed validations remain missing; 446 models were rebuilt, re-estimated and checked against their saved costs and validation errors. The primary canonical simulator evaluates the duplicate-free polynomial in straight-line Python with exactly a additions and b multiplications per sample. The secondary NumPy simulator evaluates every gene with corrected delays; it is not the unmodified package predictor. The two implementations agree within the declared reconstruction tolerances.

4.2 Timing, power integration and carbon conversion

Each model’s timing is the median of five repeated-simulation batches lasting at least 50 ms. Power was measured in a separate mains-powered session on an Apple M2 with Python 3.13.9, using CPU, GPU and neural-engine power from `powermetrics` at 100 ms intervals. Ten rounds

each visit 30 groups in a fixed seeded random order, with complete cycles over the valid models of each group for at least 10 s. Eleven idle blocks bracket the rounds, giving 311 blocks in total. Every block has valid power samples.

Net energy per simulation is

$$E_{\text{sim}} = \frac{E_{\text{block}} - \tilde{P}_{\text{idle}} t_{\text{block}}}{n_{\text{sim}}}, \quad (11)$$

where the baseline is the median power across all idle blocks. Idle block means range from 0.056 to 0.745 W. Replacing the baseline by the mean of the two idle blocks bracketing each round leaves all Holm-significance decisions unchanged in the sensitivity analysis. CodeCarbon supplies only the Irish grid intensity, 290.805 gCO₂e/kWh [17]; emissions are $E_{\text{sim}} I_{\text{grid}}$ after conversion to kWh. CodeCarbon does not track energy inside the measurement blocks and neither power nor carbon intensity establishes ρ .

Energy ratios to SO are formed within each round. Green is compared with SO, MO-lex and Green-K using two-sided Mann–Whitney tests over rounds, with Holm correction across the 18 example–simulator–comparator tests and A_{12} . These are within-session comparisons, not independent hardware sessions.

4.3 Time-based assessment of the fixed weights

For 127 distinct canonical structures, the fit $t = t_0 + t_+ a + t_\times b$ gives $t_+ = 10.16$ ns, $t_\times = 9.21$ ns and $R^2 = 0.9986$. The ratio $t_\times/t_+ = 0.906$ is close to one for this Python implementation. The operation-count fit has $R^2 = 0.9984$, compared with 0.9488 for the Karatsuba-weighted cost. These measurements support the scale of the already fixed unit weight; they do not retune the completed searches. The coefficients include interpreter overhead, are fitted to search-selected structures and do not measure instruction-level costs. The separate C microbenchmark is a diagnostic of another execution implementation.

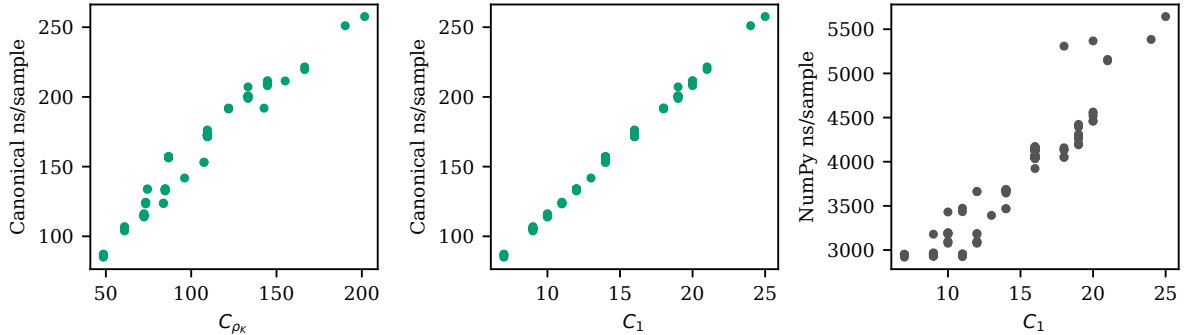


Figure 7: Corrected-model execution time per sample: canonical simulation against C_{ρ_K} and C_1 , and corrected all-gene NumPy simulation against C_1 . Both search weights are represented.

4.4 Energy results and their limits

Table 8 gives absolute time, energy and carbon estimates, while Fig. 8 compares energy relative to SO within each round. In that figure, values below one indicate savings and values above one an increase; the bars describe variation over rounds in this session. For canonical execution, median per-round Green/SO energy ratios are 0.798, 0.665 and 0.893 on E1–E3. The corresponding reductions are 20.2%, 33.5% and 10.7%. Reductions are significant against SO on E1 and E2 (Holm-adjusted $p = 0.003$ and $p = 0.003$), whereas the E3 difference is not significant

Table 8: Execution of the corrected selected models: median time, net energy and estimated carbon per simulation over ten measurement rounds. Canonical execution evaluates distinct monomials; all-gene NumPy execution uses corrected delays. Suffix K denotes searches under ρ_K .

Ex.	Method	Canonical polynomial			All-gene NumPy		
		Time (μs)	Energy (μJ)	CO ₂ e (μg)	Time (μs)	Energy (μJ)	CO ₂ e (μg)
E1	SO	9.91	65.4	0.00528	313	2.11e+03	0.17
E1	MO-lex	7.78	51.6	0.00417	329	2.31e+03	0.186
E1	Green	7.79	52.2	0.00422	337	2.37e+03	0.192
E1	MO-lex-K	7.82	52.5	0.00424	334	2.32e+03	0.187
E1	Green-K	7.78	51.1	0.00413	330	2.35e+03	0.19
E2	SO	11.7	76.7	0.0062	338	2.34e+03	0.189
E2	MO-lex	9.49	65.7	0.0053	328	2.24e+03	0.181
E2	Green	7.66	51.2	0.00414	324	2.23e+03	0.18
E2	MO-lex-K	9.44	62.7	0.00507	326	2.23e+03	0.18
E2	Green-K	7.65	50.4	0.00407	321	2.25e+03	0.182
E3	SO	38.4	251	0.0203	848	5.8e+03	0.469
E3	MO-lex	35.3	232	0.0188	832	5.55e+03	0.448
E3	Green	33.6	228	0.0184	789	5.26e+03	0.425
E3	MO-lex-K	35.4	233	0.0188	830	5.6e+03	0.452
E3	Green-K	32.9	219	0.0177	783	5.38e+03	0.434

($p = 0.542$). Green differs significantly from MO-lex in canonical energy only on E2, and not from Green-K on any example after Holm correction.

The secondary implementation changes the outcome: on E1 Green uses 11.8% more energy than SO ($p = 0.031$). E2 shows no significant difference against SO ($p = 1.000$); on E3 Green uses 7.9% less energy ($p = 0.009$), and also differs from MO-lex. Arithmetic savings of the canonical polynomial therefore do not guarantee savings when all genes are executed through NumPy. These findings concern interpreted execution on one laptop system-on-chip in one

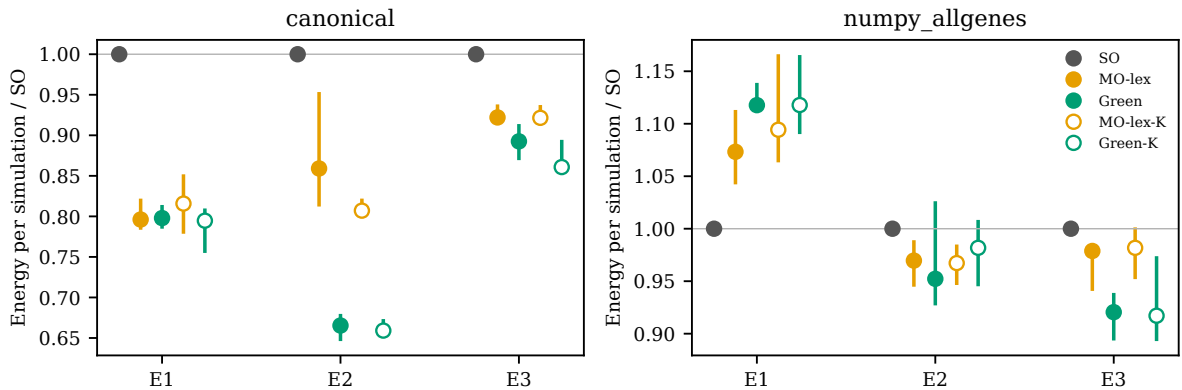


Figure 8: Corrected selected-model energy relative to SO within each of ten rounds: medians and interquartile ranges. Open markers denote the Karatsuba-weighted searches; the secondary simulator uses corrected delays.

session. They do not establish embedded-hardware performance or a universal multiplication weight. E3 selected models generalise poorly under the unchanged identification-only rule, so

energy reductions cannot be presented as savings at preserved validation accuracy. Carbon values are energy-derived estimates using one grid-intensity factor, not directly measured emissions.

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